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CPP 2202: Application Programming with VB.NET

Assignment 1 Answers and Report

1. INTRODUCTION

The aim of this assignment was to develop a Bus Ticket Booking Calculator using VB.NET Windows Forms. The application allows users to;

1. Enter passenger details
2. Select a bus route
3. Choose a passenger type and seat class
4. Specify the number of tickets
5. Select a travel date and add optional extras.

2. SYSTEM IMPLEMENTATION

The application was developed using Visual Studio and VB.NET. It provides a user-friendly interface with input controls, *Calculate, Clear* and *Exit buttons,* and a receipt display area.

The system calculates the ticket fare using *the selected route*, *passenger type, seat class* and *travel date*. It also accounts for extra luggage, travel insurance and meals. A *10% discount* is applied to bookings of five or more tickets, while a booking fee of KES 50 is added once per booking.

The program uses ***functions*** and ***procedures*** to organise the code, ***conditional statements*** to apply the pricing rules, and ***validation*** to identify incorrect or missing input. The receipt displays the fare per ticket, extras, subtotal, group discount, booking fee and final amount in Kenyan shillings.

3. TESTING AND RESULTS

The application is tested using the four compulsory test cases provided in the assignment and four additional tests covering invalid input. The expected and actual results are recorded in the table below.

Test	Input	Expected Result	Actual Result
1	Nakuru, Adult, Standard, 1 ticket, weekday, no extras	KES 750.00	PASS
2	Mombasa, Student, Business, 2 tickets, Saturday, luggage	KES 4,146.00	PASS
3	Kisumu, Child, Standard, 5 tickets, weekday, insurance and meal	KES 4,775.00	PASS
4	Eldoret, Adult, Business, 6 tickets, Sunday, all extras	KES 12,999.20	PASS
5	Passenger name left blank	Validation message	PASS
6	Passenger name contains digits	Validation message	PASS
7	Route and passenger type not selected	Relevant validation errors	PASS
8	Travel date is in the past	Validation message	PASS

Figure 1; Complete application form

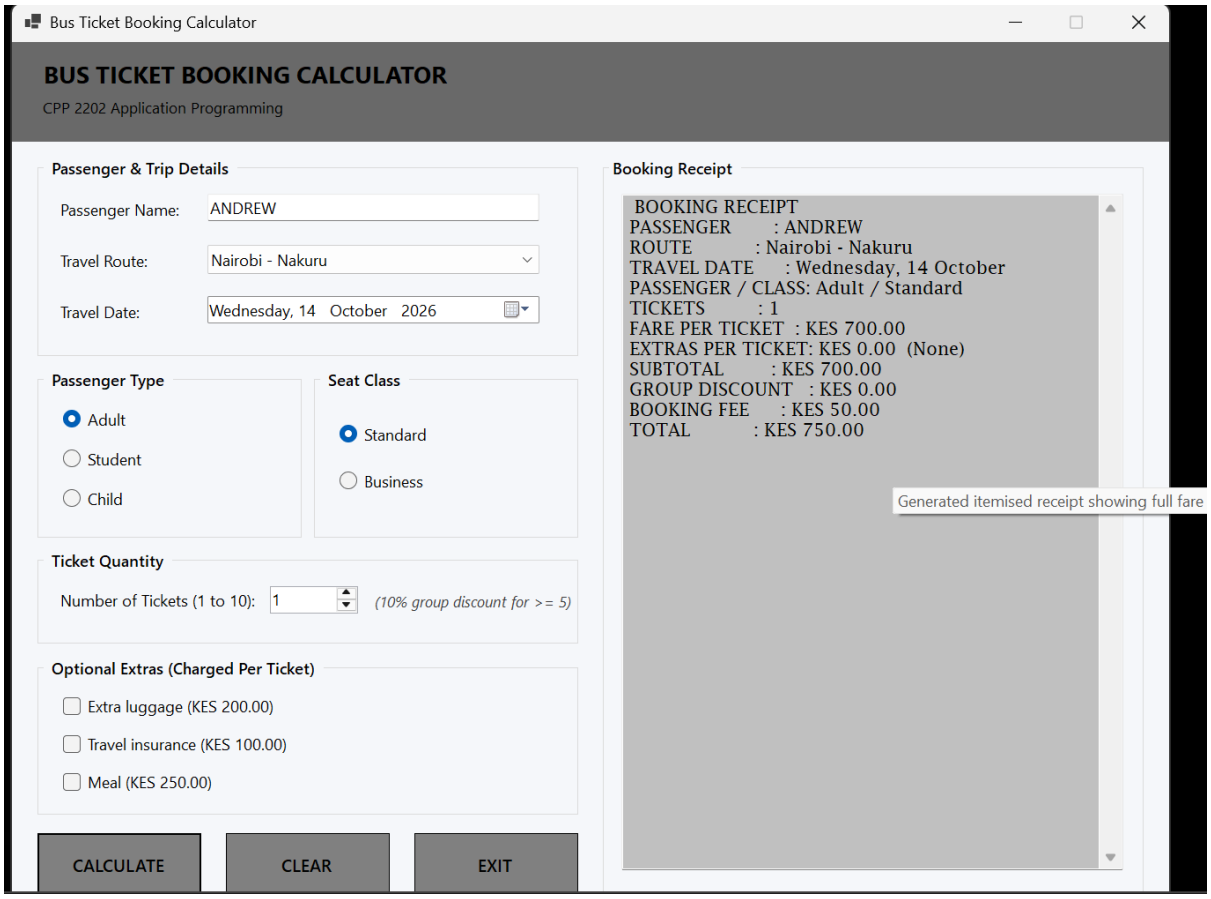


Figure 2; Successful generated receipt

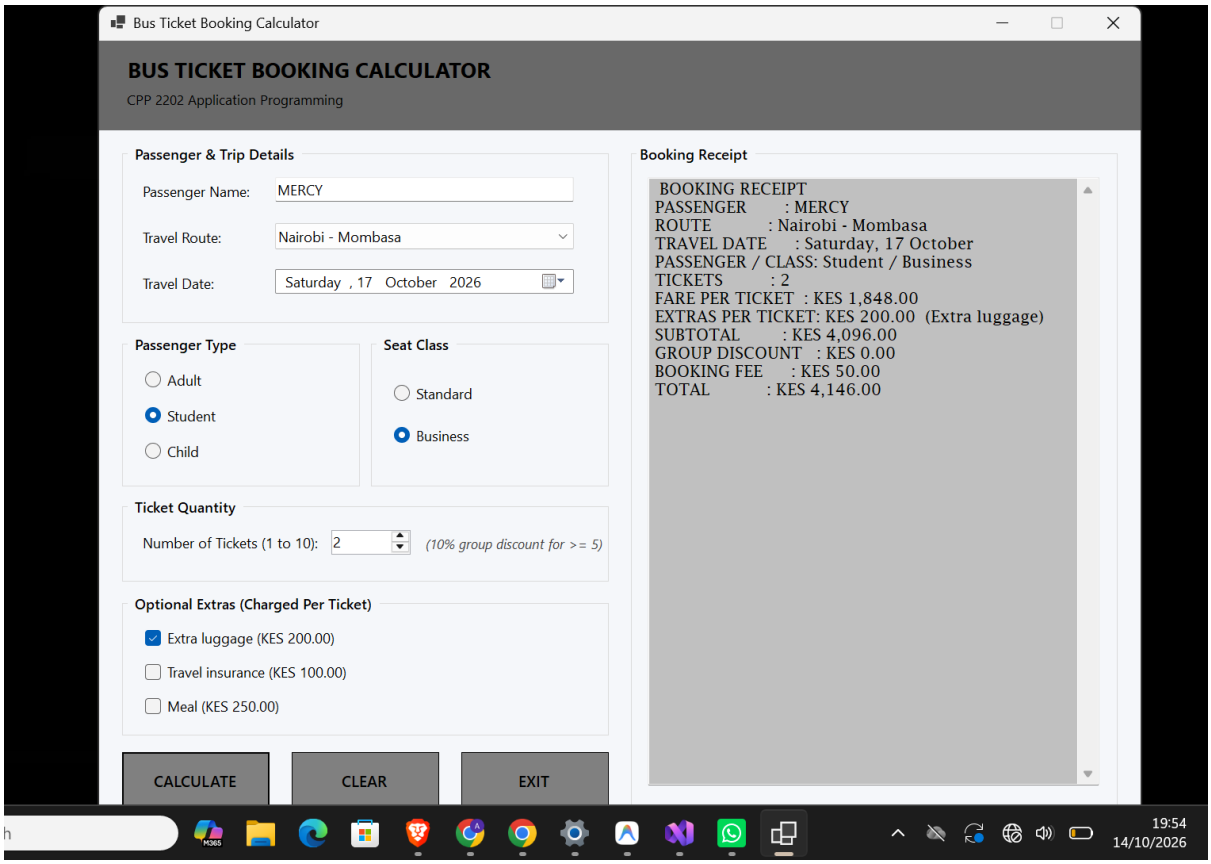


Figure 3; Validation Error Message

